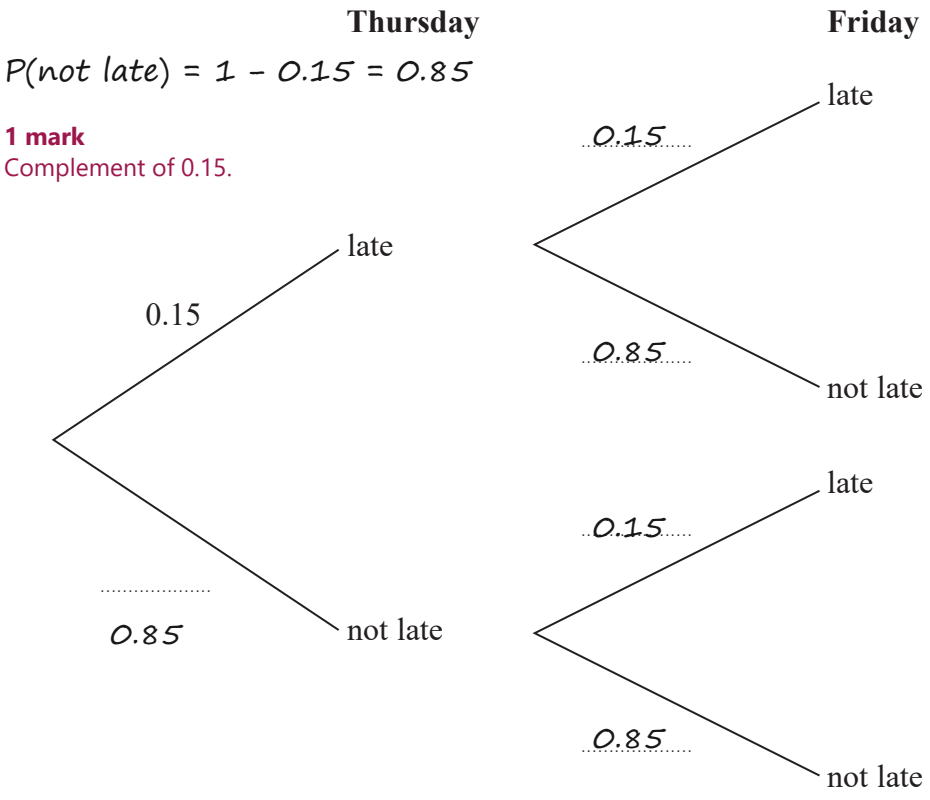
10 Mary travels to work by train every day.
The probability that her train will be late on any day is 0.15

(a) Complete the probability tree diagram for Thursday and Friday.



1 mark
Complement of 0.15.

1 mark
All branch labels correct. **(2)**

(b) Work out the probability that her train will be late on at least one of these two days.

P(late at least once) = 1 – P(no late)

= 1 – (0.85 × 0.85)

= 1 – 0.7225 = 0.2775

1 mark
Multiply one branch pair.

1 mark
Use the complement.

1 mark
0.2775.

0.2775 **(3)**

(Total for Question 10 is 5 marks)